

Tiago Carrilho Biló

Assistant Professor

02 March 1990

454 Aubert Hall, Orono, ME 04469

+1 (207)581-4335

https://orcid.org/0000-0002-4007-5862

tiago.carrilho@maine.edu

About me

I am an assistant professor at the School of Marine Sciences, University of Maine and Associated Faculty at the Climate Change Institute, University of Maine. My research interests include but are not limited to how energy, mass, and seawater properties are transported and modified by large and mesoscale processes, deep ocean circulation, ocean dynamics, and variability of the Atlantic Meridional Overturning Circulation. Although my research approach is primarily based on *in-situ* moored observations, I try to implement the best tools to answer my scientific questions.

Skills and Expertise

Physical Oceanography⋆ Ocean Currents and Circulation⋆ Data Analysis⋆ Oceanographic Instrumentation⋆ Geophysical Fluid Dynamics⋆ Coding (Python, Matlab, Fortran, LaTeX)⋆ Scientific Teaching⋆

Professional Experience

2024-Present **Assistant Professor** School of Marine Sciences, University of Maine
Main Research Topic: Large Scale Ocean Circulation

2022-2024 **Assistant Scientist** Cooperative Institute For Marine And Atmospheric Studies, University of Miami, US. Supervisor: Dr. Benjamin Kirtman
Research Topic: Deep and Abyssal Atlantic Circulation

2020-2022 **Postdoctoral Scholar** Scripps Institution of Oceanography, University of California San Diego, US. Supervisor: Dr. Fiammetta Straneo
Research Topic: Subpolar North Atlantic Circulation

Education

2015-2020 **Ph.D. student in Meteorology and Physical Oceanography**
Ph.D. Dissertation: “Pathways of the North Atlantic Deep Water in the North Atlantic subtropics: structure and recirculation dynamics".
Rosenstiel School of Marine and Atmospheric Science, University of Miami, US. Supervisor: Dr. William Johns

2013-2015 **M.Sc. student in Oceanography**
M.Sc. Thesis: “The changing Brazil Current system between 23°S-31°S: vertical structure and mesoscale dynamics"
University of São Paulo, Brazil. Supervisor: Dr. Ilson Silveira

2008-2012 **B.Sc. student in Oceanography**
B.Sc. Thesis: “On the Brazil Current System in the Santos Bifurcation area (25°S): Direct velocity measurements"
University of São Paulo, Brazil. Supervisor: Dr. Ilson Silveira

Teaching and Mentoring Programs

2025-Present **SMS400 - Capstone Research Experience in Marine Sciences** Instructor
University of Maine, US

2025-Present **SMS100 - Introduction to Ocean Sciences** Instructor
University of Maine, US

2023-2024 **CIMAS/MDC Undergraduate Internship Program** Supervisor
Supervision of intern’s 10-week-long research project over the summer at the University of Miami, US

Summer 2022 **SIO30 - The Oceans** Instructor
University of California San Diego, US

Spring 2022 **SIO90 - Undergrad Seminar and SIO-176 - Observational Physical Oceanography** Guest Lecturer
University of California San Diego, US

Spring 2021 **SIO-176 - Observational Physical Oceanography** Guest Lecturer
University of California San Diego, US

2013-2014 **Introduction to Matlab** Instructor
One week course during the 2013 and 2014 Oceanographic Week (STO) at the University of São Paulo, Brazil

Student Supervisions

Ongoing

2025-Present	Ana Clara Baptista Mariani Research Topic: Subpolar Circulation of the Deep Ocean of the North Atlantic. SMS, University of Maine, US	PhD Student
2025-Present	Andrew Treat Research Topic: Water Quality of the Maine Coast. SMS, University of Maine, US	PhD Student
2024-Present	Collin Beirne Research Topic: Vertical Mixing in the Gulf of Maine. SMS, University of Maine, US	MSc Student
2026-Present	Jay Broccolo Research Topic: Impact of a Weakening AMOC on the U.S. NE Coast. CCI, University of Maine, US	PhD Student
2026-Present	Marina Aguiar Research Topic: South Atlantic Abyssal Circulation. University of Sao Paulo, Brazil	Visiting PhD Student
2024-Present	Zoe Crow Research Topic: Deep Circulation of the Gulf of Maine. SMS, University of Maine, US	Oceanography Student (undergraduate)
2024-Present	Nova Johnson Research Topic: Abyssal Atlantic Equatorial Circulation. SMS, University of Maine, US	Oceanography Student (undergraduate)
2026-Present	Teddy Kelley Research Topic: Nitrate Variability in the Gulf of Maine. SMS, University of Maine, US	Oceanography Student (undergraduate)

Concluded

2024-2026	Ella Boxall Research Topic: Sea Level Variability in Bar Harbor, ME. MCEC, University of Maine, US	Civil Engineering Student (undergraduate)

Field Experience

OSNAP Project	Research Cruises DY182, AR102-01 Lead moorings and CTD/LADCP operations	Chief Scientist: Tiago Carrilho Bilo
OSNAP Project	Research Cruises AR30-04 and DY053 Perform moorings and CTD/LADCP operations; salinity water sampling and onboard analysis	Chief Scientists: Dr. William Johns and Dr. Stuart Cunningham
MOCHA Project	Research Cruises AE1833, and AB1705 Perform moorings and CTD/LADCP operations	Chief Scientist: Dr. William Johns
CERES Experiment	Research Cruises CERES V (leg 3) and CERES IV Instruments (XBT, CTD, SADCP, and LADCP) operations, near-real time data processing, and adaptive sampling	Chief Scientist: Dr. Ilson Carlos A. da Silveira and Dr. Wellington Ceccopieri Belo
AMBES Experiment	Research Cruise AMB09 Near-real time data processing and adaptive sampling	Chief Scientist: Dr. Ilson Carlos A. da Silveira

Awards

NSF #1948198	OSNAP: $3,078,524.00 https://www.nsf.gov/awardsearch/showAward?AWD_ID=1948198	Co-PIs: Dr. William Johns and Dr. Tiago Carrilho Bilo
NOAA-NERACOOS: Year 5	$100,000.00 https://neracoos.org/projects/nutrient-observatory/	UMaine PI: Tiago Carrilho Bilo
CO_2 Foundation:	$54,939 https://co2foundation.org/modeling-and-sharing-nonlinear-climate-risks-in-the-northeastern-u-s/	UMaine PI: Tiago Carrilho Bilo

Publications

Articles in Journals

Published

1. Foukal, N.; Le Bras, I; Fu, Y; Petit T; Biló, T C; Elipot, S; Moat, B, 2026: “Strengthening connections in observing the North Atlantic Meridional Overturning Circulation: Outcomes from a joint RAPID-OSNAP workshop", Oceanography, doi: 10.5670/oceanog.2026.e110
2. Santos, D. M. C.; Biló, T C; Napolitano, D C; Perez, R C; Polito, P S; Gula, J; Dong, S; Campos, E J D; Sato, O T, 2026: “Antarctic bottom water contraction drives abyssal ocean warming along SAMBA-West line (34.5°S) in the Argentine basin", Deep-Sea Research Part I, doi: 10.1016/j.dsr.2025.104627
3. Fu, Y.; Lozier, S M; Burmeister, K; Biló, T C; et al., 2025: “Characterizing the interannual variability of North Atlantic subpolar overturning", Geophysical Research Letters, doi: 10.1029/2025GL114672
4. Fried, N.; Biló, T C; Johns, W; Katsman, C A; de Jong, F M, 2024: “Freshening over the whole water column as a result of the 2012 subpolar freshwater anomaly increased the transport of lighter waters of the Irminger Current between 2014-2022", Journal of Geophysical Research: Oceans, doi: 10.1029/2024JC021184
5. Koman, G.; Bower, A A; Holliday, N P; Furey, H H; Fu, Y; Biló, T C, 2024: “The decreasing transport of the Deep Western Boundary Current in the subpolar North Atlantic", Nature Geoscience, doi: 10.1038/s41561-024-01555-6
6. Biló, T. C; Perez, R; Dong, S; Johns, W E; Kanzow, T, 2024: “Weakening of the Atlantic Meridional Overturning Circulation Abyssal Limb in the North Atlantic", Nature Geoscience, doi: 10.1038/s41561-024-01422-4
7. Amorin, J. P.; Silveira, I C A; Borges-Silva, M; Souza-Neto, P W M; Bernardo, P S; Dottori, M; Belo, W C; Martins, R P; Biló, T C, 2024: “Brazil Current Cyclonic Meandering at 27°S-31°S: the Santa Marta Eddies Variability and Biogeochemical Responses", Deep Sea Research Part I: Oceanographic Research Papers, doi: 10.1016/j.dsr.2024.104276
8. Johnson, G. C.; Lumpkin, R; Atkinson, C; Biló, T; et al., 2023: "Global Oceans", Bulletin of the American Meteorological Society, doi: 10.1175/BAMS-D-23-0076.2
9. Fu, Y.; Lozier, S M; Biló, T C; et al., 2023: “Seasonality of the Meridional Overturning Circulation in the Subpolar North Atlantic", Communications Earth & Environment, doi: 10.1038/s43247-023-00848-9
10. Vilela-Silva, F.; Silveira, I C A; Napolitano, D C; Souza-Neto, P W M; Biló, T C; Gangopadhyay, A, 2023: “On the Deep Western Boundary Current Separation and Anticyclone Genesis off Northeast Brazil", Journal of Geophysical Research: Oceans, doi: 10.1029/2022JC019168
11. Le Bras, I. A.-A.; Callies, J; Straneo, F; Biló, T C; Holte, J; Johnson, H L, 2022: “Slantwise convection in the Irminger Sea", Journal of Geophysical Research: Oceans, doi: 10.1029/2022JC019071
12. Biló, T. C.; Straneo, F; Holte, J; Le Bras, I A-A, 2022: “Arrival of New Great Salinity Anomaly Weakens Convection in the Irminger Sea", Geophysical Research Letters, doi: 10.1029/2022GL098857
13. Desbruyeres, D. G.; Bravo, E P; Thierry, V; Mercier, H; Pascale, L; Cécile, C; Biló, T C; Fried, N; De Jong, M F, 2022: “Warming-to-Cooling Reversal of Overflow-Derived Water Masses in the Irminger Sea During 2002–2021", Geophysical Research Letters, doi: 10.1029/2022GL098057
14. Biló, T. C.; Johns, W E; Zhao, J., 2021: “Dynamics of deep recirculation cells offshore of the Deep Western Boundary Current in the subtropical North Atlantic (15°-30°N)", Journal of Physical Oceanography, doi: 10.1175/JPO-D-20-0184.1
15. Biló, T. C.; Johns, W E, 2020: “The Deep Western Boundary Current and adjacent interior circulation at 24-30°N: Mean structure and mesoscale variability", Journal of Physical Oceanography, doi: 10.1175/JPO-D-20-0094.1
16. Biló, T. C.; Johns, W E, 2019: “Interior pathways of Labrador Sea Water in the North Atlantic from the Argo perspective", Geophysical Research Letters, doi: 10.1029/2018GL081439
17. Biló, T. C.; da Silveira, I C A; Belo, W C; Castro, B M; Piola, A R, 2014: “Methods for estimating the velocities of the Brazil Current in the pre-salt reservoir area off southeast Brazil (23°S-26°S)", Ocean Dynamics, doi: 10.1007/s10236-014-0761-2

Book Chapters

1. Soares, S. M.; Steck, V; Biló, T C; Chen, C; 2026: “Intertwined: How Ocean Ecosystems and Climate Interact". In D. Ebert-May & E. Holt (Eds.), Pathways to scientific teaching, Volume 2. Macmillan Higher Education. VitalSource Bookshelf.

Datasets and Codes

1. Fu, Y.; Lozier S. M.; Bower A.; Burmeister K.; Biló T. C.; et al., 2025: “Meridional Overturning Circulation Observed by the Overturning in the Subpolar North Atlantic Program (OSNAP) Array from August 2014 to July 2024" [Dataset], doi: 10.35090/gatech/78023

2. Kanzow, T.; Biló, T. C.; Lankhorst, M.; Mauritzen, C., 2023: “Guyana Abyssal Gyre Experiment (GAGE) current meter mooring ocean velocity records obtained between February 2000 and April 2002 at 16°N" [Data set], doi: 10.17604/5jd9-7f77
3. Fu, Y.; Lozier S. M.; Biló T. C.; et al., 2023: “Meridional Overturning Circulation Observed by the Overturning in the Subpolar North Atlantic Program (OSNAP) Array from August 2014 to June 2020" [Dataset], doi: 10.35090/gatech/70342
4. Straneo, F.; Torres, D.; Bahr, F.; Holte, J.; Biló, T. C., 2021: “Water temperature, salinity, and velocity collected by CTD, ADCP, and current meters from OSNAP East moorings CF1 - CF7 in the Irminger Sea from 2016-08-16 to 2020-07-17" [Data set], NCEI Accession 0226020, NOAA National Centers for Environmental Information, https://www.ncei.noaa.gov/archiv
5. Biló, T. C. 2019: “North Atlantic Observed Climatological Mean Absolute Geostrophic Velocity Profiles" [Data set], doi: 10.17604/cf5z-x124

Peer-review contributions

I acted as a reviewer for several journals and institutions, including, but not limited to, the National Science Foundation, Nature Geoscience, Nature Climate Change, Geophysical Research Letters, Journal of Physical Oceanography, Progress in Oceanography, Deep Sea Research, and Journal of Geophysical Research: Oceans, Philosophical Transactions of the Royal Society A.

Conferences Attended as Presenter

Oral Presentations

1. Biló, T. C; Perez, R; Dong, S; et al., 2023: “Weakening of the Atlantic Meridional Overturning Circulation Abyssal Limb in the North Atlantic". In: AGU Fall Meeting, 2023, San Francisco, USA.
2. Biló, T. C; Perez, R; Dong, S; et al., 2023: “Weakening of the Atlantic Meridional Overturning Circulation Abyssal Limb in the North Atlantic". In: Workshop on Meeting AMOC Observation Needs in a Changing Climate, 2023, Germany.
3. Biló, T. C., Straneo, F; Holte, J, 2022: “The East Greenland Coastal Current near Cape Farewell". In: Arctic-Subarctic Ocean Fluxes, 2022, Iceland.
4. Biló, T. C., Straneo, F; Holte, J; Le Bras, I A-A, 2022: “On the Recent Freshening of the Irminger Sea (2015-2020)". In: Ocean Sciences Meeting, 2022, Virtual Conference.
5. Biló, T. C., Johns, W. E., 2019: “Interior pathways of Labrador Sea Water in the North Atlantic from the Argo perspective". In: EGU General Assembly, 2019, Vienna.

Poster Presentations

1. Biló, T. C; Johns, W. E.; Mercier, H.; Lherminier, P.; Desbruyères, D., 2026: “A decade of East Reykjanes Ridge Current observations: velocity structure and transport variability". In: Ocean Sciences Meeting, 2026, Glasgow, UK.
2. Biló, T. C; Perez, R; Dong, S; et al., 2024: “Weakening of the Atlantic Meridional Overturning Circulation Abyssal Limb in the North Atlantic". In: Ocean Sciences Meeting, 2024, New Orleans, USA.
3. Biló, T. C., Johns, W. E., 2020: “Deep Western Boundary Current variability at 26.5°N and its connection with local abyssal recirculation.". In: Ocean Sciences Meeting, 2020, San Diego.
4. Biló, T. C., Johns, W. E., 2018: “On the mean Deep Western Boundary Current recirculation in the North Atlantic subtropics: structure and forcing mechanisms". In: Ocean Sciences Meeting, 2018, Portland.
5. Biló, T. C., Silveira, I. C. A., Rocha, C. B., Belo, W. C., 2014: “ON THE BRAZIL CURRENT THICKENING IN SANTOS BASIN (23-28°S)". In: Ocean Sciences Meeting, 2014, Honolulu.
6. Biló, T. C.; Paloczy, A., Rocha, C. B., Silveira, I. C. A., Ceccopieri, W., 2012: “ON THE BRAZIL CURRENT SYSTEM OFF SOUTHEAST BRAZIL (22°S): TOP - BOTTOM DIRECT VELOCITY MEASUREMENTS". In: Ocean Sciences Meeting, 2012, Salt Lake City.